VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Microcontroller

Lab Report

Lab 1: LED Animations

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1 Exercise 1

Report 1: Schematic

The Proteus schematic for this exercise is shown below.

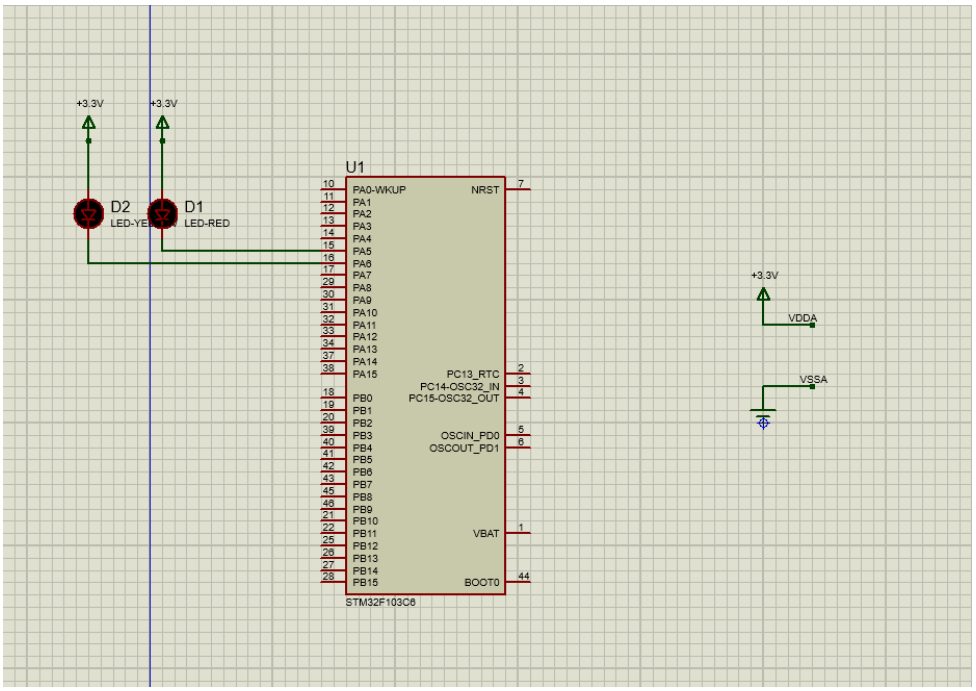


Figure 1.1: https://github.com/aulequangtri/microprocessor_lab

Report 2: Source Code

The source code inside the `while(1)` loop to switch the states of the RED and YELLOW LEDs every 2 seconds:

```
while (1)
{
    HAL_GPIO_TogglePin(LED_RED_GPIO_Port, LED_RED_Pin);
    HAL_GPIO_TogglePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin);
    HAL_Delay(2000);
}
```

2 Exercise 2

Report 1: Schematic

The Proteus schematic for the 3-LED traffic light system is shown below.

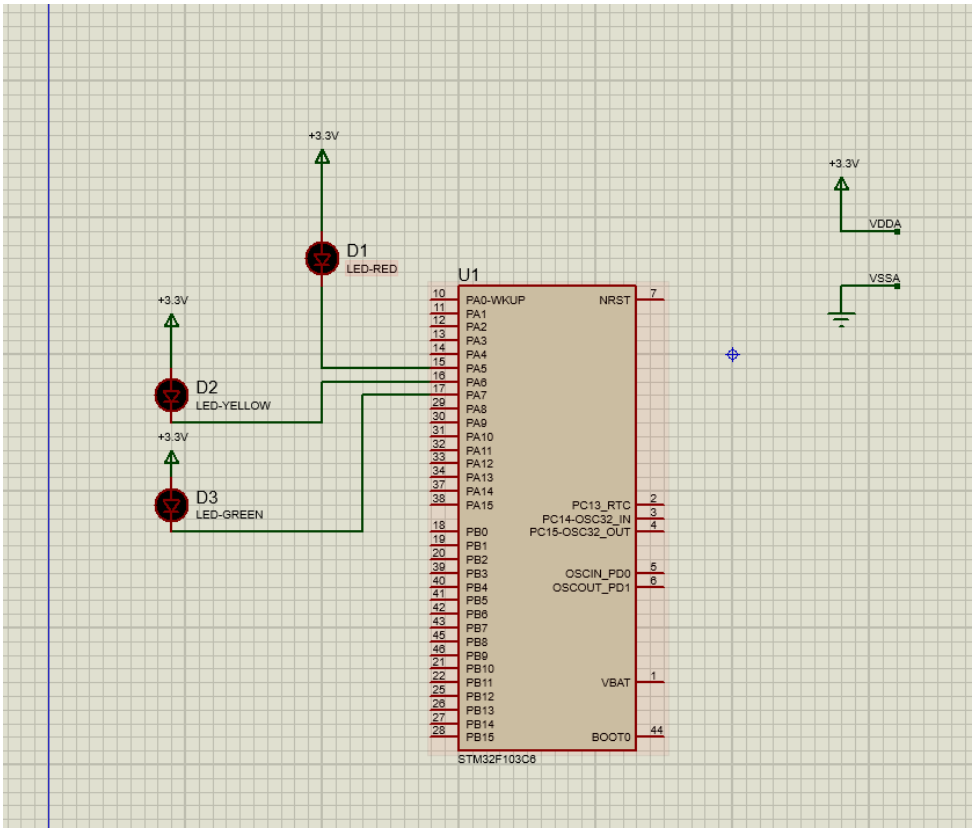


Figure 2.1: Schematic for Exercise 2 (Traffic Light)

Report 2: Source Code

The source code inside the `while(1)` loop simulating the traffic light cycle:

```
while (1)
{
    HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, RESET);
    HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, SET);
    HAL_Delay(5000);

```

```
    HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, SET);
    HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, SET);
    HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, RESET);
    HAL_Delay(3000);

    HAL_GPIO_WritePin(LED_RED_GPIO_Port, LED_RED_Pin, SET);
    HAL_GPIO_WritePin(LED_YELLOW_GPIO_Port, LED_YELLOW_Pin, RESET);
    HAL_GPIO_WritePin(LED_GREEN_GPIO_Port, LED_GREEN_Pin, SET);
    HAL_Delay(2000);
}
```

3 Exercise 4

Report 1: Schematic

The schematic for the 7-segment display integration.

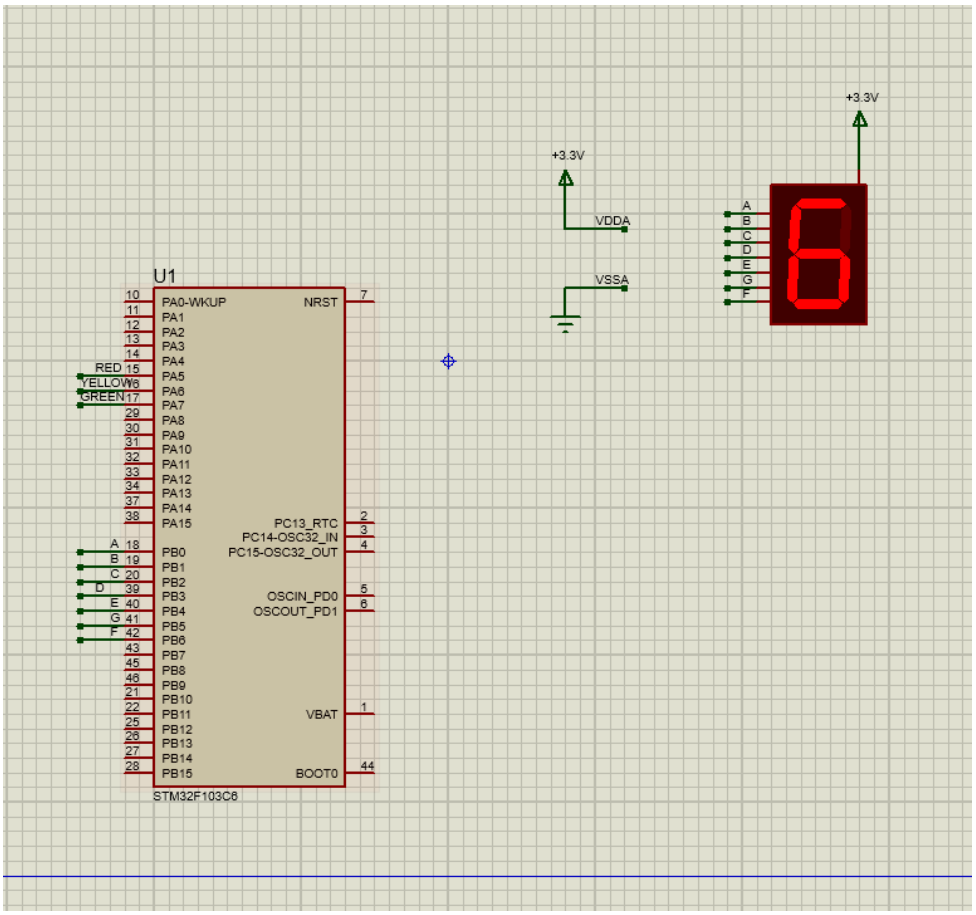


Figure 3.1: Schematic for Exercise 4 (7-Segment LED)

Report 2: Source Code for display7SEG function

The function `display7SEG(int num)` is implemented as follows:

```
void display7SEG(int num) {
      /* Turn off all segments first */
      HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin, GPIO_PIN_SET);
      HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin, GPIO_PIN_SET);

```



```
11      /* Turn on required segments based on the number */
12      switch (num) {
13          case 0:
14              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
15              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
16              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
17              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
18              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
                GPIO_PIN_RESET);
19              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                GPIO_PIN_RESET);
20              break;
21          case 1:
22              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
23              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
24              break;
25          case 2:
26              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
27              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
28              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
29              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
                GPIO_PIN_RESET);
30              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                GPIO_PIN_RESET);
31              break;
32          case 3:
```

```
33              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
34              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
35              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
36              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
37              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                GPIO_PIN_RESET);
38              break;
39          case 4:
40              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
41              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
42              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                GPIO_PIN_RESET);
43              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                GPIO_PIN_RESET);
44              break;
45          case 5:
46              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
47              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
48              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
49              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                GPIO_PIN_RESET);
50              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                GPIO_PIN_RESET);
51              break;
52          case 6:
```



```
53              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
54              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
55              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
56              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
                GPIO_PIN_RESET);
57              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                GPIO_PIN_RESET);
58              HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                GPIO_PIN_RESET);
59              break;
60          case 7:
61              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
62              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
63              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
64              break;
65          case 8:
66              HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                GPIO_PIN_RESET);
67              HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                GPIO_PIN_RESET);
68              HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                GPIO_PIN_RESET);
69              HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                GPIO_PIN_RESET);
70              HAL_GPIO_WritePin(LED_E_GPIO_Port, LED_E_Pin,
                GPIO_PIN_RESET);
71              HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                GPIO_PIN_RESET);
```



```
72                  HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                    GPIO_PIN_RESET);
73                  break;
74              case 9:
75                  HAL_GPIO_WritePin(LED_A_GPIO_Port, LED_A_Pin,
                    GPIO_PIN_RESET);
76                  HAL_GPIO_WritePin(LED_B_GPIO_Port, LED_B_Pin,
                    GPIO_PIN_RESET);
77                  HAL_GPIO_WritePin(LED_C_GPIO_Port, LED_C_Pin,
                    GPIO_PIN_RESET);
78                  HAL_GPIO_WritePin(LED_D_GPIO_Port, LED_D_Pin,
                    GPIO_PIN_RESET);
79                  HAL_GPIO_WritePin(LED_F_GPIO_Port, LED_F_Pin,
                    GPIO_PIN_RESET);
80                  HAL_GPIO_WritePin(LED_G_GPIO_Port, LED_G_Pin,
                    GPIO_PIN_RESET);
81                  break;
82          }
83  }
```

4 Exercise 5

Report 1: Schematic

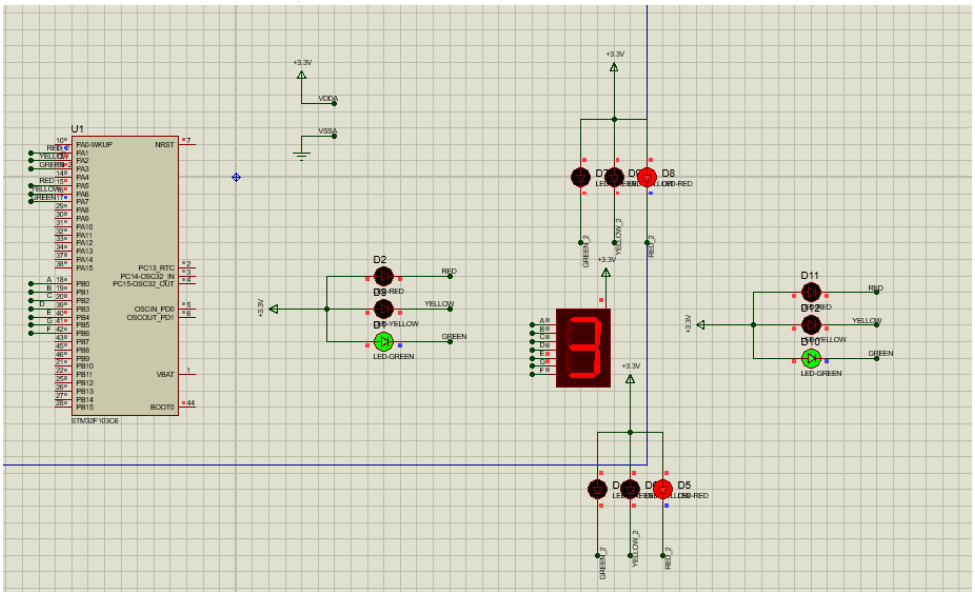


Figure 4.1: Schematic for Exercise 5 (4-Way Traffic Light + 7SEG)

Report 2: Source Code

The code snippet inside the infinite loop for the 4-way traffic light integrated with a 7-segment countdown:

```
while (1)
{
    display7SEG(countdown);

    HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(YELLOW_GPIO_Port, YELLOW_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(GREEN_GPIO_Port, GREEN_Pin, GPIO_PIN_SET);

    HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(YELLOW_2_GPIO_Port, YELLOW_2_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(GREEN_2_GPIO_Port, GREEN_2_Pin, GPIO_PIN_SET);

```



```
13      if (state == 0) {
14          HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_RESET);
15          HAL_GPIO_WritePin(GREEN_2_GPIO_Port, GREEN_2_Pin,
            GPIO_PIN_RESET);
16
17          countdown--;
18          if (countdown <= 0) {
19              state = 1;
20              countdown = 2;
21          }
22      }
23      else if (state == 1) {
24          HAL_GPIO_WritePin(RED_GPIO_Port, RED_Pin, GPIO_PIN_RESET);
25          HAL_GPIO_WritePin(YELLOW_2_GPIO_Port, YELLOW_2_Pin,
            GPIO_PIN_RESET);
26
27          countdown--;
28          if (countdown <= 0) {
29              state = 2;
30              countdown = 3;
31          }
32      }
33      else if (state == 2) {
34          HAL_GPIO_WritePin(GREEN_GPIO_Port, GREEN_Pin, GPIO_PIN_RESET);
35          HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_RESET);
36
37          countdown--;
38          if (countdown <= 0) {
39              state = 3;
40              countdown = 2;
41          }
42      }
43      else if (state == 3) {
44          HAL_GPIO_WritePin(YELLOW_GPIO_Port, YELLOW_Pin,
            GPIO_PIN_RESET);
```

```
45          HAL_GPIO_WritePin(RED_2_GPIO_Port, RED_2_Pin, GPIO_PIN_RESET);
46
47          countdown--;
48          if (countdown <= 0) {
49              state = 0;
50              countdown = 3;
51          }
52      }
53
54      HAL_Delay(1000);
55  }
```

5 Exercise 6

Report 1: Schematic

The Proteus schematic for the analog clock using 12 LEDs.

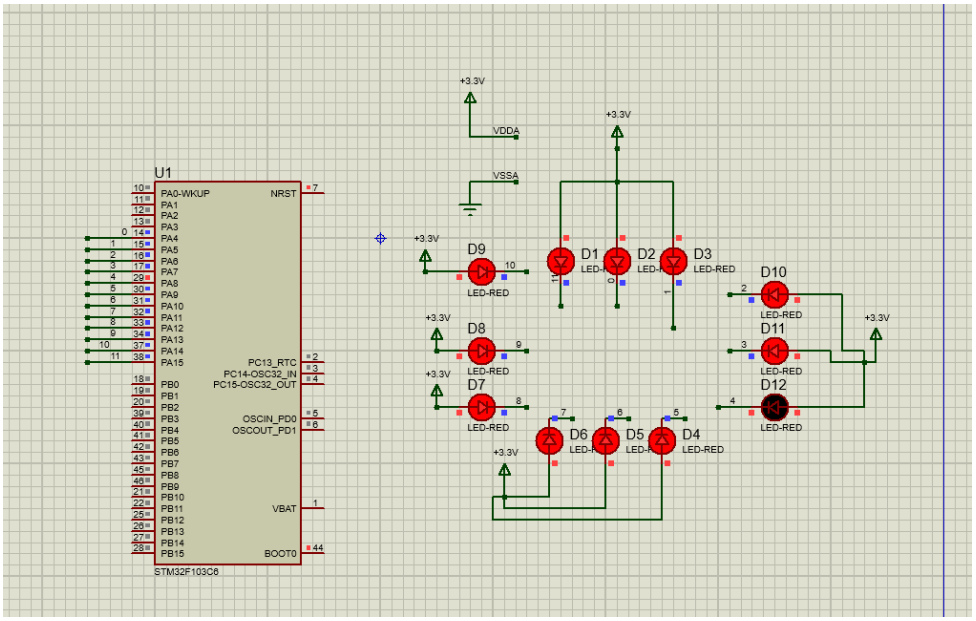


Figure 5.1: Schematic for Exercise 6 (Analog Clock)

Report 2: Source Code

The testing program turning every LED in a sequence:

```
while (1)
{
    HAL_GPIO_WritePin(GPIOA, clock_pins[current_led], GPIO_PIN_SET);

    current_led++;

    if (current_led >= 12) {
        HAL_Delay(1000);

        for (int i = 0; i < 12; i++) {
            HAL_GPIO_WritePin(GPIOA, clock_pins[i], GPIO_PIN_RESET);
        }
        current_led = 0;
    }

    HAL_Delay(500);
}
```

6 Exercise 7 to 10

The following schematic is used for the analog clock exercises (7 to 10):

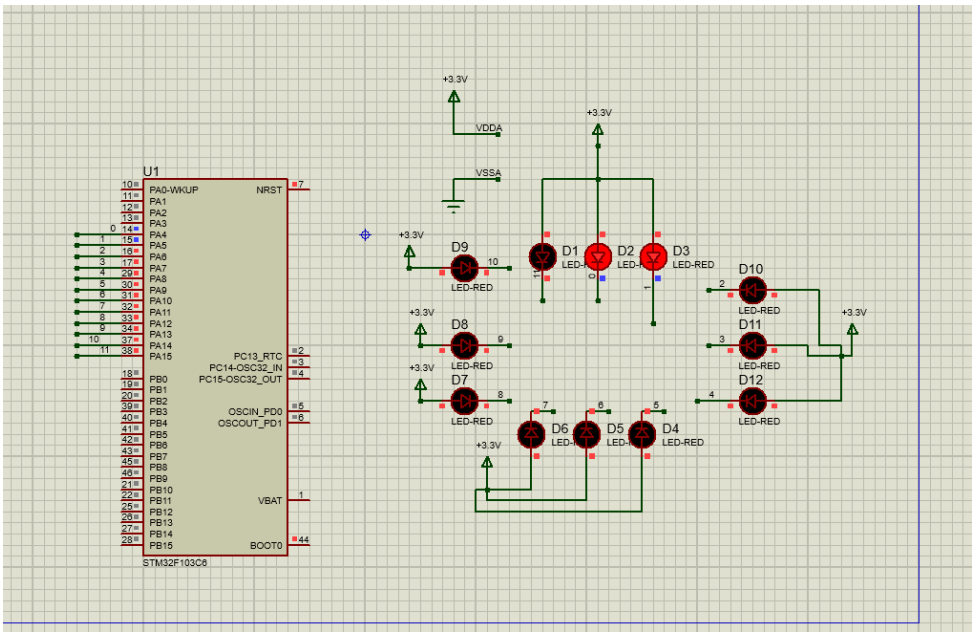


Figure 6.1: Schematic for Exercises 7-10 (Analog Clock Functions)

Exercise 7: clearAllClock()

```
1  void clearAllClock(void) {
2      uint16_t all_clock_pins = GPIO_PIN_4 | GPIO_PIN_5 | GPIO_PIN_6 |
       GPIO_PIN_7 |
3                                GPIO_PIN_8 | GPIO_PIN_9 | GPIO_PIN_10 |
                                 GPIO_PIN_11 |
4                                GPIO_PIN_12 | GPIO_PIN_13 | GPIO_PIN_14
                                 | GPIO_PIN_15;
5
6      HAL_GPIO_WritePin(GPIOA, all_clock_pins, LED_OFF);
7  }
```

Exercise 8: setNumberOnClock(int num)

```
1  void setNumberOnClock(int num) {
2      if (num >= 0 && num <= 11) {
3          uint16_t clock_pins[12] = {
4              GPIO_PIN_4, GPIO_PIN_5, GPIO_PIN_6,  GPIO_PIN_7,
```

```
            GPIO_PIN_8, GPIO_PIN_9, GPIO_PIN_10, GPIO_PIN_11,
            GPIO_PIN_12, GPIO_PIN_13, GPIO_PIN_14, GPIO_PIN_15
        };
        HAL_GPIO_WritePin(GPIOA, clock_pins[num], LED_ON);
    }
}
```

Exercise 9: clearNumberOnClock(int num)

```
void clearNumberOnClock(int num) {
    if (num >= 0 && num <= 11) {
        uint16_t clock_pins[12] = {
            GPIO_PIN_4,  GPIO_PIN_5,  GPIO_PIN_6,  GPIO_PIN_7,
            GPIO_PIN_8,  GPIO_PIN_9,  GPIO_PIN_10, GPIO_PIN_11,
            GPIO_PIN_12, GPIO_PIN_13, GPIO_PIN_14, GPIO_PIN_15
        };
        HAL_GPIO_WritePin(GPIOA, clock_pins[num], LED_OFF);
    }
}
```

Exercise 10: Analog Clock Main Loop

The integration of the system using 12 LEDs to display an analog clock (Hour, Minute, and Second).

```
while (1) {
    clearAllClock();

    int hour_led = hour % 12;
    int min_led = minute / 5;
    int sec_led = second / 5;

    setNumberOnClock(hour_led);
    setNumberOnClock(min_led);
    setNumberOnClock(sec_led);
```

```
11
12      HAL_Delay(1000);
13
14      second++;
15      if (second >= 60) {
16          second = 0;
17          minute++;
18
19          if (minute >= 60) {
20              minute = 0;
21              hour++;
22
23              if (hour >= 12) {
24                  hour = 0;
25              }
26          }
27      }
28  }
```